

Quadratic Equations: Derivation and Discriminant

Standard form: $ax^2 + bx + c = 0$, $a \neq 0$

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminant $D = b^2 - 4ac$

$D > 0$: two distinct real roots

$D = 0$: two equal real roots

$D < 0$: no real roots